Quantum Exposure Report

capgemini.com Public TLS endpoint · port 443 Scanned 14 Sep 2026

MIGRATION REQUIRED

QUANTUM RISK	ASSETS ANALYSED	QUANTUM-VULNERABLE	IMMEDIATE ACTION
ELEVATED	3	2	2
weighted across assets	1 protocol · 2 primitives	of 3 catalogued	critical priority

CRYPTOGRAPHIC BILL OF MATERIALS

ASSET · ROLE	QUANTUM STATUS	PRIORITY	RECOMMENDED
TLS 1.3 Transport protocol	Safe	LOW	Protocol current
ECDSA-P256 Certificate signature	Vulnerable	CRITICAL	ML-DSA (FIPS 204)
X25519 Key exchange	Vulnerable	CRITICAL	ML-KEM-768 (FIPS 203)

CONNECTION EVIDENCE

TLS version	TLS 1.3	Certificate	ECDSA-P256
Cipher suite	CHACHA20-POLY1305	Issuer	Let's Encrypt
Key exchange	X25519	Expires	2026-10-16
Forward secrecy	Enabled	Origin / IP	Direct · 192.0.66.152

WHY IT MATTERS

Both primitives protecting this endpoint — the certificate signature (**ECDSA-P256**) and the session key exchange (**X25519**) — rest on elliptic-curve mathematics that a cryptographically-relevant quantum computer can break. Traffic captured today can be stored and decrypted later once that capability exists — the **harvest-now, decrypt-later** risk. The transport protocol itself (TLS 1.3) is current; the exposure is in the algorithms it carries.

RECOMMENDED REMEDIATION

1 **Deploy a hybrid key exchange — X25519MLKEM768**

Closes the harvest-now exposure on the session key immediately, while staying interoperable with classical clients. Supported on OpenSSL 3.5+ / modern TLS stacks.

2 **Plan the certificate move to ML-DSA-65**

Track CA and PKI support for FIPS 204 signatures; stage a hybrid certificate chain as issuers enable it.

REGULATORY CONTEXT — INDIA

PCI DSS 12.3.3 · SEBI CSCRF · RBI Q-SAFE · DST Task Force · 2027 · DPDP Act 2023

Tool seqr-scanner 0.1.0 · SeQr Labs

Format CycloneDX 1.6 CBOM

Scan method Passive TLS · 6.4s

Confidence High (direct origin, no CDN)

Passive-scan note. Findings reflect the cryptography this server advertised at scan time. A passive scan cannot confirm the *absence* of hybrid post-quantum support; an active ML-KEM handshake probe settles that.

HNDL disclaimer. Harvest-now-decrypt-later exposure scores are predictive cryptographic risk metrics based on data-type endurance, not indicators of active data interception or confirmed network breaches.